

Expanded octets and VSEPR (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- Represent species with five and six electron domains around a central atom using Lewis formulas.
- Deduce the electron domain geometry and the molecular geometry for these species using the VSEPR model.

Phosphorus trichloride, PCl_3 , is an important industrial chemical with many uses, including in the production of pesticides. Use information from [section S2.2.4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/vsepr-id-45129/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/vsepr-id-45129/>) to help you draw the Lewis formula for PCl_3 .

Phosphorus pentachloride, PCl_5 , is another industrially important molecule used in the production of pesticides. The Lewis formula for PCl_5 is given for you in **Figure 1**.

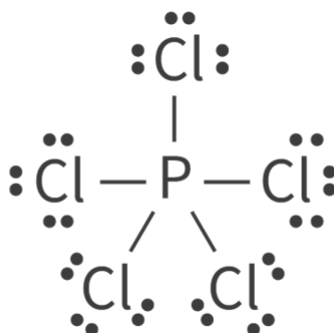


Figure 1. Lewis formula of phosphorus pentafluoride, PCl_5 , with an expanded octet.

 More information for figure 1

This image displays the Lewis structure of phosphorus pentachloride (PCl_5). Central to the diagram is a phosphorus atom (P), which is surrounded symmetrically by five chlorine atoms (Cl), each connected to the phosphorus by a single bond. The chlorine atoms are distributed above, below, and on the sides of the phosphorus atom, forming a trigonal bipyramidal shape. Each

chlorine atom is represented with its respective lone pairs of electrons depicted as two dots next to the Cl symbol. The structure exemplifies an expanded octet for the phosphorus atom, with ten electrons around it, accommodating the five chlorine atoms. This arrangement is possible due to phosphorus's ability to expand its valence shell to include more than eight electrons.

[Generated by AI]

How is it possible for phosphorus to be bonded to five chlorine atoms at the same time?

Violations of the octet rule

The octet rule is one that is largely followed by most atoms, but there are some exceptions. Section S2.2.1a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/>) describes the formation of molecules that contain atoms with incomplete octets; atoms that can also be described as electron-deficient.

It is also possible for atoms to violate the octet rule by having more than eight electrons in their outer shell, as phosphorous in PCl_5 shows. This is known as having an expanded octet (**Figure 2**).

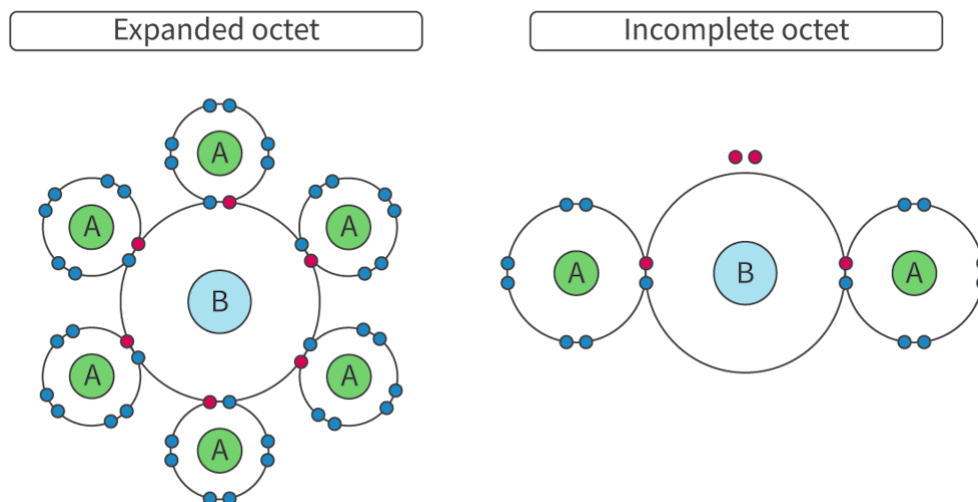


Figure 2. Expanded and incomplete octets both violate the octet rule.

 More information for figure 2

This image contains two diagrams comparing expanded and incomplete octets. On the left is the expanded octet. In this diagram, a central atom labeled 'B' is surrounded by five smaller atoms labeled 'A'. Each 'A' atom has a ring of blue dots representing electrons, and the central 'B' atom has pink dots connecting it to each 'A'. The arrangement shows more than eight electrons surrounding the central atom, demonstrating an expanded octet.

On the right is the incomplete octet diagram. Here, a central atom 'B' has electron pairs represented by dots but fewer connections than in the expanded octet. Two 'A' atoms are positioned around 'B', each with fewer electrons, illustrating an incomplete octet with less than eight electrons around the central atom.

These diagrams visually depict the concept of electron distribution in chemical structures, illustrating how atoms can have expanded or incomplete electron shells.

[Generated by AI]

Achieving stability with an incomplete octet is typically only possible for smaller atoms and can be linked to their position on the periodic table; anything beyond period 2 is unlikely to be stable with fewer than eight electrons in their outer shell.

The ability of a central atom to achieve stability with an expanded octet can also be linked to its position on the periodic table; it is only possible for an element in period 3 onwards. Why is this the case? These atoms have empty 3d-orbitals which, in terms of energy, are very close to the 3p-orbitals. This makes it possible to promote electrons from the 3p to the 3d-orbitals if needed, which allows additional electron pairs to form. Why is this not possible for elements in period 2 or below?

Take phosphorus pentafluoride, PF_5 , as an example (**Figure 3**). In total, it has 40 valence electrons and so the only way the molecule can form is if phosphorus expands its octet to hold 10 electrons.

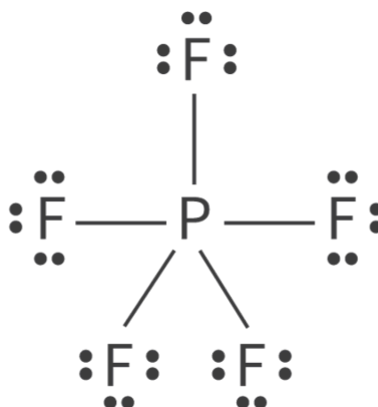


Figure 3. Lewis formula of phosphorus pentafluoride, PF_5 , with an expanded octet.

 More information for figure 3

The image depicts the Lewis structure of phosphorus pentafluoride (PF_5). In the center, there is a phosphorus atom, represented by 'P'. The phosphorus atom is surrounded by five fluorine atoms, denoted by 'F'. Each fluorine atom is connected to the phosphorus atom via a single line, symbolizing a single covalent bond. Each fluorine atom has three pairs of dots around it,

representing the non-bonding electron pairs. The phosphorus atom, in this expanded octet structure, does not show electron pairs around it in this representation.

[Generated by AI]

To determine whether a molecule forms with an expanded octet you should follow the same rules to drawing a Lewis formula, which is covered in [section S2.2.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/multiple-covalent-bonds-id-45128/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/multiple-covalent-bonds-id-45128/>).

Drawing Lewis formulas for molecules and ions that have central atoms with an expanded octet

1. Calculate the total number of valence electrons in the molecules. Adjust for the charge if required.
2. Identify the central atom. Draw the basic structure of the molecule by linking the peripheral atoms to the central atom with single bonds.
3. Add remaining electrons (following the octet rule).
4. If there are not enough electrons to complete the octets, form double and triple bonds if necessary.
5. Check that the total number of electrons in the finished structure is equal to your calculation in the first step.
6. If all peripheral atoms have fulfilled the octet rule and there are still electrons remaining, add them as lone pairs to the central atom, provided the central atom is from period 3 or beyond.

Worked example 1

Draw the Lewis formula for

1. XeF₂
2. SF₆

1. XeF₂

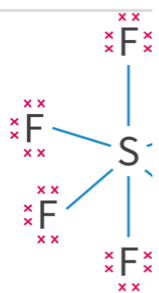
Solution steps	Calculations
Step 1: Calculate total valence electrons.	$\text{Xe} + (\text{F} \times 2) = 8 + (7 \times 2)$ $= 22 \text{ total}$

Solution steps	Calculations
Step 2: Identify the central atom. Here, it will be xenon as it is the least numerous (only 1 xenon compared to 2 fluorine). Join the peripheral atoms (fluorine) to the central atom (xenon) with single bonds.	$\text{F} \text{ --- } \text{Xe} \text{ ---}$
Step 3: Add remaining electrons as valence electrons.	$\begin{array}{c} \times \times \\ \times \text{F} \text{ --- } \text{Xe} \text{ ---} \\ \times \times \end{array} \quad \begin{array}{c} \times \times \\ \times \times \end{array}$
Step 4: Add double or triple bonds where necessary to ensure that all atoms fulfil octet rule. (Octet rule already achieved so not necessary.)	
Step 5: Check the number of electrons added is correct as calculated in Step 1. No, 20 electrons are added in total.	
Step 6: As there is no space for additional electrons on	

Solution steps	Calculations
peripheral atoms, they must be added on the central atom. Xenon is in period 5 so it can accommodate these additional electrons and expand its octet.	

2. SF₆

Solution steps	Calculations
Step 1: Calculate total valence electrons.	$S + (F \times 6) = 6 + (7 \times 6)$ $= 48 \text{ total}$
Step 2: Identify the central atom. Here, it will be sulfur as it is the least numerous (only 1 sulfur compared to 6 fluorine). Join the peripheral atoms (fluorine) to the central atom (sulfur) with single bonds.	There are 6 fluorine atoms that must be in period 3, so it can expand its octet: the s 
Step 3: Add remaining electrons as valence electrons.	

Solution steps	Calculations
	
Step 4: Add double or triple bonds where necessary to ensure that all atoms fulfil octet rule. (Octet rule already achieved so not necessary.)	
Step 5: Check the number of electrons added is as calculated in Step 1.	Yes, 48 electrons added in total.
Step 6: Octet already expanded for sulfur.	

VSEPR and expanded octets

Section S2.2.4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/vsepr-id-45129/>) shows how the molecular geometry could be determined by identifying the number and type of electron domains surrounding a central atom in a molecule. The standard level course shows how to do this for species that:

1. fulfils the octet rule
2. violates the octet rule by being electron-deficient (has an incomplete octet).

As a higher-level student, you should also be able to determine the molecular geometry of molecules with expanded octets, species with up to six electron domains.

Species with five electron domains

A species with five electron domains will either have trigonal bipyramidal, see-saw, T-shaped or linear geometry, depending on the relative number of non-bonding pairs and bonding pairs.

Before we can understand where these different shapes originate from, it can be helpful to understand where the bonding and non-bonding pairs of electrons are located around the central atom.

Figure 4 shows that there are two different regions of electron density in molecules: the axial and the equatorial. Both of the axial bonds lay along the y-axis whereas the three equatorial bonds are found on the equator of the molecule.

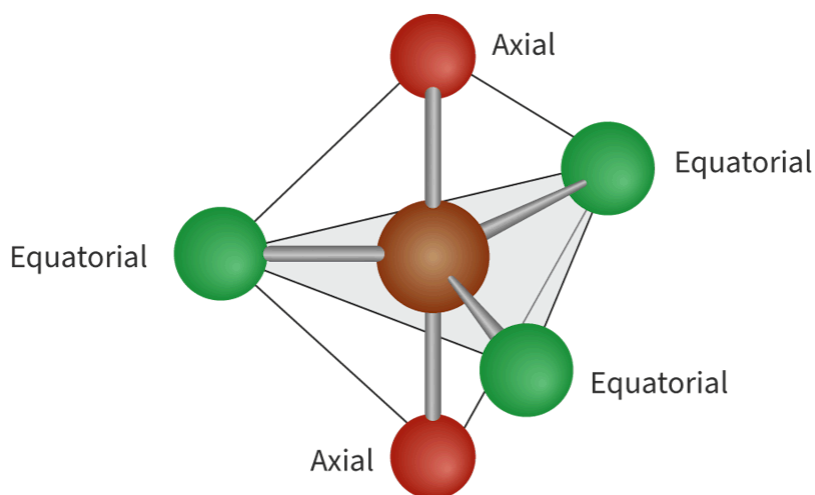


Figure 4. Axial atoms lay on the vertical plane while equatorial atoms lay on the horizontal plane.

 More information for figure 4

The image is a diagram depicting a molecular structure with atoms arranged in specific positions. At the center is a central atom shown in brown. Two red atoms labeled "Axial" are positioned above and below this central atom, forming a vertical line along the y-axis. Three green atoms labeled "Equatorial" are located around the central atom, forming a triangle or equatorial plane. These equatorial atoms are spaced evenly around the central atom, representing the plane perpendicular to the axial line. Thin lines connect the central atom to the axial and equatorial atoms, illustrating the bonds between them.

[Generated by AI]

Section S2.2.4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/vsepr-id-45129/>) covers electron domain geometries.

What shape is formed when two electron domains are present?

Linear

What shape is formed when three electron domains are present?

Trigonal planar

When five electron domains are present, and there are no non-bonding pairs, the molecular geometry formed is trigonal bipyramidal (**Figure 5**). Why do we end up with this shape?

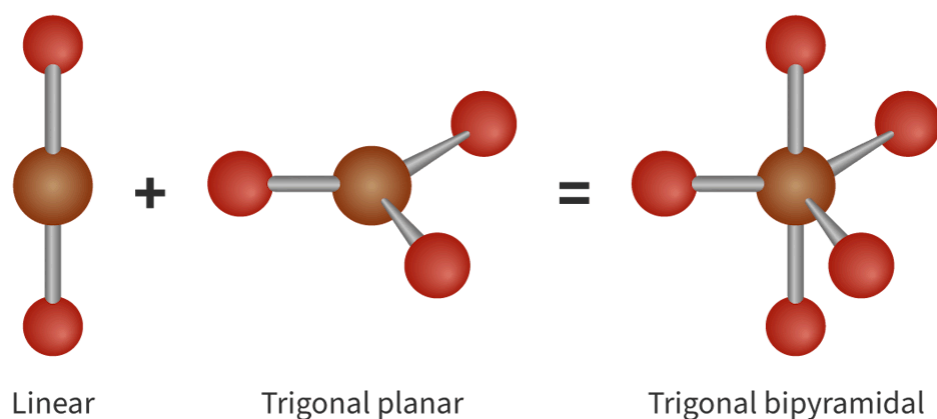


Figure 5. Trigonal bipyramidal.

 More information for figure 5

The image displays three molecular structures. On the left, there is a linear structure labeled "Linear" with a central brown atom connected to two red atoms in a straight line. In the middle, a trigonal planar structure is depicted, labeled "Trigonal planar," with a central brown atom connected to three red atoms in a triangular arrangement. On the right, the structures combine to form a "Trigonal bipyramidal" shape. This final structure has a central brown atom connected to five red atoms: two directly opposite each other (linear), and three forming a triangular plane around the central atom. The combination signifies the transition from simpler linear and trigonal planar arrangements to a more complex trigonal bipyramidal geometry.

[Generated by AI]

To help you visualise this shape, try to picture two three-sided pyramids that have their bases touching, as in **Figure 6**. Can you now see why it's called bipyramidal?

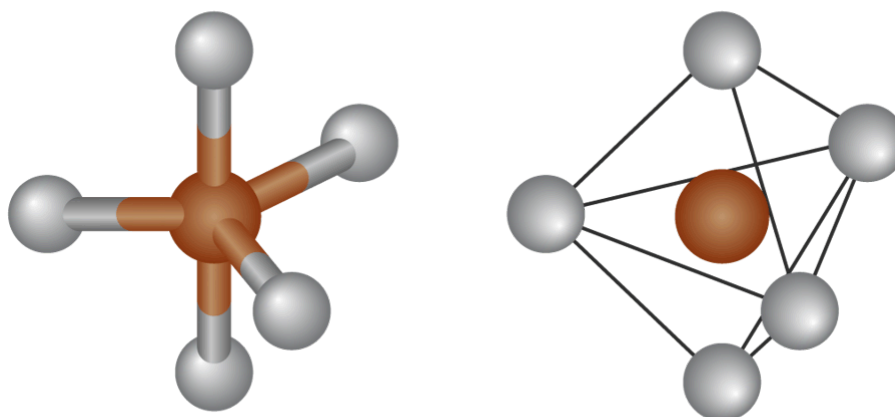


Figure 6. Trigonal bipyramidal.

 More information for figure 6

The image shows a diagram representing trigonal bipyramidal molecular geometry. On the left, there is a 3D model with a central atom connected to five peripheral atoms. Three peripheral atoms are in a plane, equidistant from each other, forming a triangle (equatorial positions). The other two are positioned above and below this plane (axial positions), completing the bipyramidal shape. On the right, a wireframe model depicts the same arrangement with lines connecting the atoms to illustrate the geometric shape. This configuration demonstrates the spatial arrangement where equatorial atoms have bond angles of 120° between them, and axial atoms have bond angles of 90° with equatorial atoms, forming 180° between themselves.

[Generated by AI]

As **Figure 7** shows, the bond angles between the equatorial atoms are 120° , between the linear atoms is 180° and the angles between the equatorial and axial atoms are 90° .

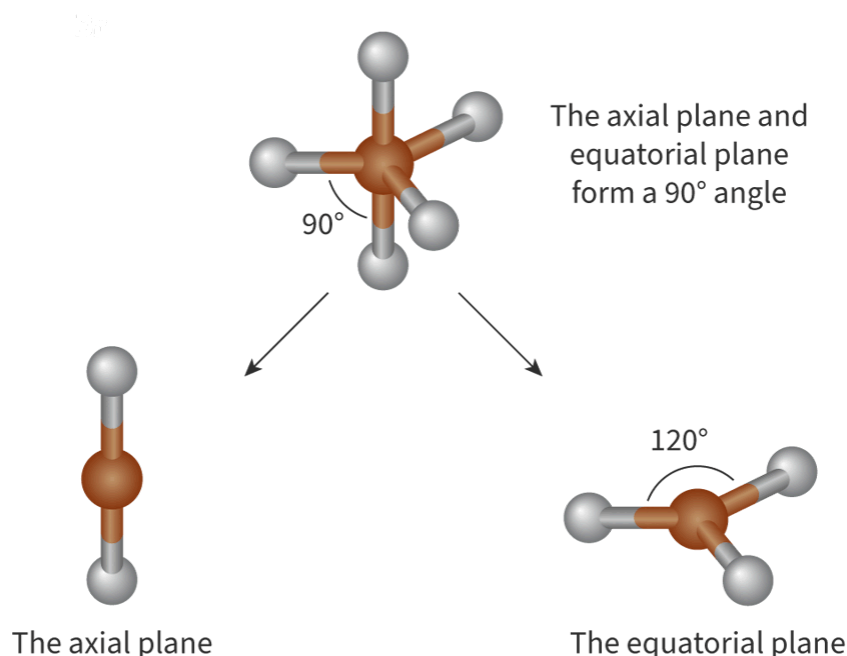


Figure 7. Trigonal bipyramidal.

 More information for figure 7

The image depicts the trigonal bipyramidal molecular geometry. It shows a central atom (bromine) bonded to five surrounding atoms positioned at the vertices of a trigonal bipyramid. The geometry is separated into two planes: the axial and equatorial planes. The axial plane consists of one axis with two atoms bonded at 180° from each other. The equatorial plane has three atoms bonded at 120° angles from one another, forming a horizontal plane perpendicular to the axial plane. The angles formed within this molecular structure include 90° between the axial and equatorial planes. The diagram visually explains these angles with arrows and accompanying text annotations, outlining the significance of the geometric arrangement in molecular chemistry.

[Generated by AI]

Non-bonding pairs cause more repulsion than bonding pairs, so when non-bonding pairs are added the molecular geometry becomes a little distorted.

One non-bonding pair of electrons and four bonding pairs results in a see-saw molecular geometry, which you can see in **Figure 8**. This can also be known as sawhorse molecular geometry.

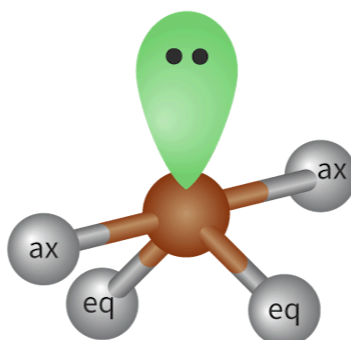


Figure 8. See-saw shape.

 More information for figure 8

The image is a diagram depicting the see-saw molecular geometry, also known as sawhorse geometry. The structure includes a central atom represented by a bronze sphere, with four surrounding atoms colored silver, each attached to the central atom by brown bonds. These atoms are labeled as 'ax' and 'eq', denoting axial and equatorial positions respectively. Additionally, a larger green, teardrop-shaped lobe is present, representing a lone pair of electrons. This arrangement illustrates the asymmetric shape of the see-saw molecular geometry, where two axial atoms are opposite each other, and two equatorial atoms are adjacent but not directly opposite each other.

[Generated by AI]

Can you see where the name 'see-saw' comes from?

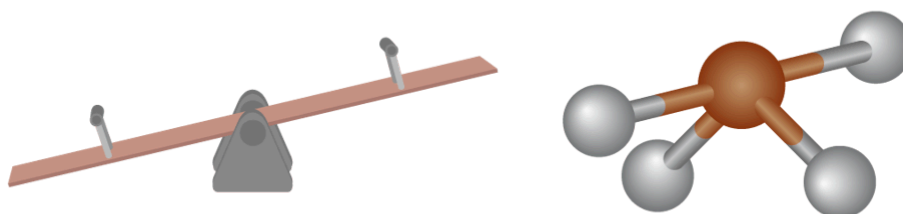


Figure 9. See-saw shape.

 More information for figure 9

This image shows two illustrations side by side. On the left, there is a see-saw, which consists of a long plank balanced in the center, allowing it to tilt up and down. Two small figures represent positions where people might sit. On the right, there is a diagram representing a trigonal bipyramidal molecule shape. It has a central atom connected to five surrounding atoms: three in a plane and two above and below, resembling the appearance of a see-saw. The see-saw and the molecular shape are juxtaposed to highlight their structural similarity, specifically illustrating the concept of equatorial and axial positions as found in certain chemical structures.

[Generated by AI]

As **Figure 9** shows, similar to the trigonal bipyramidal molecule, the see-saw also has equatorial and axial regions.

The non-bonding pair of electrons, located in the equatorial position, will cause additional repulsion. This causes some bond angles to be slightly less than that of the trigonal bipyramidal; the axial-equatorial angle will be less than 90° (e.g. approximately 87°), and the equatorial bond angles will be less than 120° (e.g. approximately 117°). The axial bond angle is $<180^\circ$ (**Figure 10**).

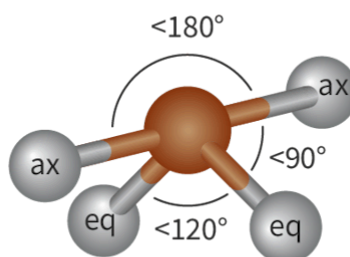


Figure 10. See-saw bond angles.

 More information for figure 10

The diagram illustrates a see-saw molecular geometry showing bond angles in a trigonal bipyramidal arrangement. There is a central atom connected to four peripheral atoms. Two bond angles are specifically labeled as less than 180 degrees, less than 90 degrees, and less than 120 degrees. These peripheral atoms are noted as positioned axially (labeled 'ax') and equatorially (labeled 'eq'). The labeling indicates positions and their corresponding bond angles: axial-equatorial angles less than 90 degrees, equatorial-equatorial angles less than 120 degrees, and the axial-axial angle less than 180 degrees. The diagram is focused on visualizing electron repulsion effects and resultant bond angle modifications in this molecular structure.

[Generated by AI]

A **T-shaped** geometry occurs when there are three bonded pairs of electrons and two non-bonding pairs. Can you see from **Figure 11** why it is called T-shaped?

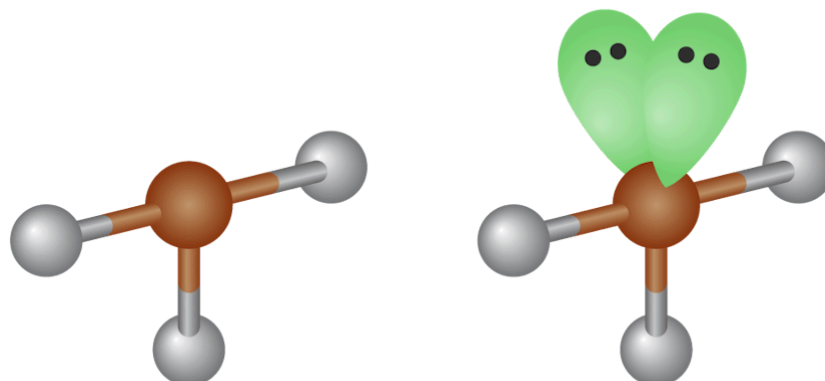


Figure 11. T-shaped.

 More information for figure 11

The image depicts models of T-shaped molecular geometry. On the left, a central atom is shown with three bonded electron pairs extending in a T formation. On the right, the same structure is augmented with two non-bonding electron pairs depicted as green lobes above the central atom. The geometry explains the presence of three bonded pairs such that two pairs form a linear arrangement at 180° in the horizontal plane, while the third bonded pair is perpendicular to these in the vertical plane. The non-bonding pairs occupy positions where equatorial bonds would typically be, resulting in a less than 90° angle due to additional repulsion.

[Generated by AI]

The two non-bonding pairs sit where two of the equatorial bonds usually would, and so the linear bond angle is still 180° . The bond angle between the linear and equatorial atom is less than 90° (e.g. approximately 88°) as the non-bonding pairs provide additional repulsion (**Figure 12**).

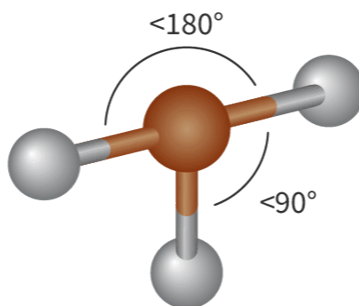


Figure 12. T-shaped bond angles.

 More information for figure 12

The image is a 3D molecular model illustrating T-shaped bond angles. It features a central atom connected to three peripheral atoms, forming a T shape. Two bond angles are labeled: one as ' $<180^\circ$ ', indicating a linear bond, and the other as ' $<90^\circ$ ', highlighting the angle between the linear and equatorial atoms. This visual representation aids in understanding the spatial arrangement of electron pairs and the resulting bond angles due to electron repulsion, illustrating the geometry described in surrounding text.

[Generated by AI]

The final geometry for a molecule with five electron domains occurs when there are two bonded pairs of electrons and three non-bonding pairs. The three non-bonding pairs are all found on the equatorial region and the resulting molecular geometry is a **linear** geometry (**Figure 13**).

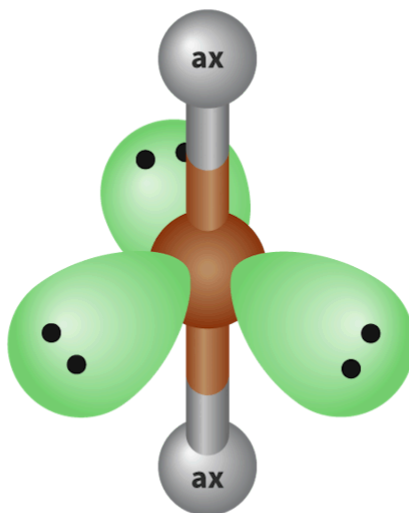
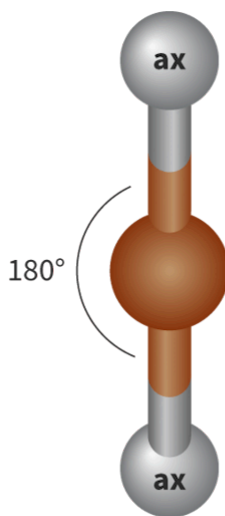


Figure 13. Linear geometry. More information for figure 13

The diagram depicts a linear molecular geometry with a central brown atom connected to two linear gray rods labeled "ax", which stand for axial positions. These rods are connected to gray spheres representing the axial atoms. Three green lobes are drawn around the central atom in a trigonal manner, indicating non-bonding electron pairs in the equatorial plane. Each lobe has two small black dots, symbolizing electrons. This configuration represents a molecule with five electron domains: two bonded pairs (the axial atoms) and three non-bonding pairs (the equatorial lobes). The overall geometry is linear with a bond angle of 180° between the bonded pairs.

[Generated by AI]

This looks indistinguishable from the other linear geometries that occur when a molecule has two electron domains. The bond angle between the two bonded pairs is 180° (**Figure 14**).

**Figure 14. Linear bond angles.** More information for figure 14

The image is a diagram illustrating a linear molecular geometry with two bonded pairs positioned 180° apart. At the center is a brown sphere representing the central atom. At each end, there are smaller gray spheres labeled 'ax,' indicating axial atoms. A 180° angle is marked between the two axial atoms. This visual represents a straight-line structure where the central atom is connected to two peripheral atoms, forming a 180° -degree bond angle, typical of a linear electron domain geometry.

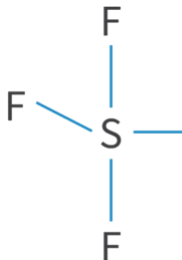
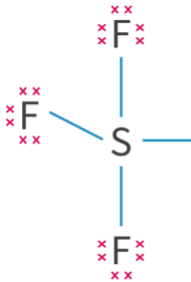
[Generated by AI]

Worked example 2

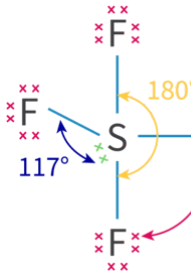
Draw the Lewis formula and identify the molecular geometry and predict the bond angles for:

1. Sulfur tetrafluoride, SF₄
2. Chlorine trifluoride, ClF₃
3. Triiodide ion, I₃[−]

1. Sulfur tetrafluoride, SF₄

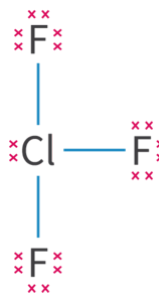
Solution steps	Calculations
Step 1: Calculate total valence electrons.	$\text{S} + (\text{F} \times 4) = 6 + (7 \times 4)$ $= 34 \text{ total}$
Step 2: Identify the central atom. Here it will be sulfur as it is the least numerous (only 1 sulfur compared to 4 fluorine). Join the peripheral atoms (fluorine) to the central atom (sulfur) with single bonds.	
Step 3: Add remaining electrons as valence electrons.	

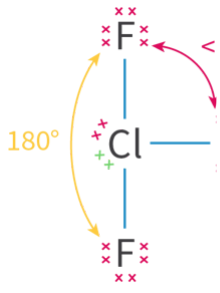
Solution steps	Calculations
Step 4: Add double or triple bonds where necessary to ensure that all atoms fulfil octet rule. (Octet rule already achieved so not necessary.)	
Step 5: Check the number of electrons added is correct as calculated in Step 1. No, only 32 electrons are added in total.	
Step 6: Octet must expand for sulfur. This is possible, as it's in period 3.	
Step 7: Identify molecular geometry and bond angles.	Electron domains: 5 Bonded electron pairs: 4 Lone electron pairs: 1 Molecular geometry: See-saw Predicted angles: 90° , 180° and $<120^\circ$ (e.g. 117°)

Solution steps	Calculations
	

2. Chlorine trifluoride, ClF₃

Solution steps	Calculations
Step 1: Calculate total valence electrons.	$\text{Cl} + (\text{F} \times 3) = 7 + (7 \times 3)$ $= 28 \text{ total}$
Step 2: Identify the central atom. Here it will be chlorine as it is the least numerous (only 1 chlorine compared to 3 fluorine). Join the peripheral atoms (fluorine) to the central atom (chlorine) with single bonds.	

Solution steps	Calculations
Step 3: Add remaining electrons as valence electrons.	
Step 4: Add double or triple bonds where necessary to ensure that all atoms fulfil octet rule. (Octet rule already achieved so not necessary.)	
Step 5: Check the number of electrons added is correct as calculated in Step 1. No, 26 electrons are added in total.	

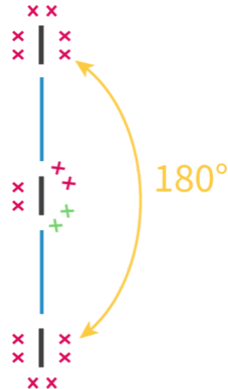
Solution steps	Calculations
Step 6: Octet must expand for chlorine. This is possible as it's in period 3.	
Step 7: Identify molecular geometry and bond angles.	Electron domains: 5 Bonded electron pairs: 3 Lone electron pairs: 2 Molecular geometry: T-shaped Angles: $<90^\circ$ and 180° 

3. Triiodide ion, I_3^-

Solution steps	Calculations
Step 1: Calculate total valence electrons.	$(\text{I} \times 3) + (\text{negative charge}) = (7 \times 3) + (1) = 22$

Solution steps	Calculations
Step 2: Identify the central atom. Here it will be one of the iodines, as it is the only option. Join the peripheral atoms (other iodines) to the central atom (iodine) with single bonds.	
Step 3: Add the remaining electrons as valence electrons.	
Step 4: Add double or triple bonds where necessary to ensure that	

Solution steps	Calculations
all atoms fulfil octet rule. (Octet rule already achieved so not necessary.)	
Step 5: Check the number of electrons added is correct as calculated in Step 1. No, only 20 electrons are added in total.	
Step 6: Octet must expand for iodine. This is possible as it's in period 3	
Step 7: Identify shape and bond angles.	Electron domains: 5 Bonded electron pairs: 2 Lone electron pairs: 3 Molecular geometry: Linear Angles: 180°

Solution steps	Calculations
	

In each of the examples before, bond angles are predicted based on the VSEPR theory. These are estimates of the angles likely to be found but are not precise values. For example, in sulfur tetrafluoride, SF_4 , the predicted bond angles are 90° , 180° and 117° whereas the actual bond angles are 89° , 173° and 102° . In your exams, you are required to be able to **predict** values, not state exact values.

Species with six electron domains

Species with six electron domains will also have different molecular geometries depending on the number of non-bonding electron pairs. They will either have octahedral, square pyramidal or square planar molecular geometry.

If there are six bonded pairs and no non-bonding pairs present, an octahedral molecular geometry will occur.

To help you visualise this in 3D and understand where the name has come from, first think of a pyramid that has four sides. Now imagine two of those, placed with their bases touching. This would form an octahedron with eight sides, as you can see in **Figure 15**. Can you see why the name 'octahedral' is given to this molecular geometry?

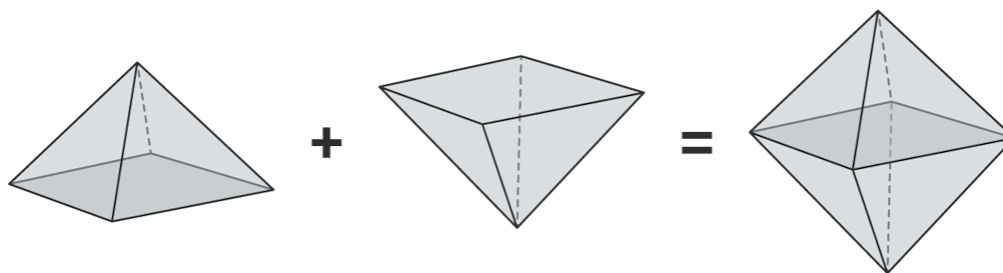


Figure 15. Octahedron.

 [More information for figure 15](#)

The image illustrates the formation of an octahedron by combining two pyramids. On the left, there is a pyramid with four triangular faces. Next to it, on the right, is a similar pyramid oriented upside-down. These two pyramids are shown with their bases touching, demonstrating how together they form an octahedron, which has eight triangular faces in total. The image uses a plus sign and an equals sign to indicate the addition process, resulting in the octahedron on the right. Dotted lines indicate the axis of symmetry for each shape.

[Generated by AI]

What will be the bond angle associated with this geometry? Analyse **Figure 16** to work this out.

You may find it helpful to add a square to the electron pairs to help you identify the angle.

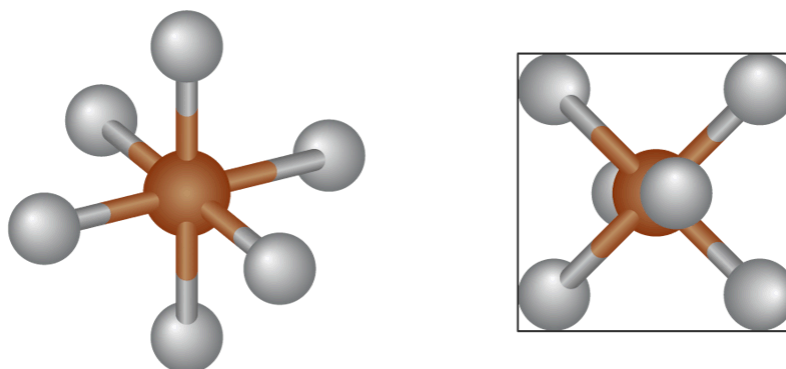


Figure 16. Octahedral.

 [More information for figure 16](#)

The image displays two 3D models depicting octahedral molecular geometry. On the left, a central atom is connected to six surrounding atoms through lines representing bonds. These surrounding atoms are all equidistant from the center and placed symmetrically to form an octahedral shape. On the right, a similar model is shown from a different angle, illustrating the geometric configuration from a top-down perspective.

[Generated by AI]

The octahedral shape will only have 90° bond angles.

When a non-bonding pair of electrons is present, the additional repulsion they cause will alter the bond angle. Five bonded pairs and one non-bonding pair of electrons results in a square pyramidal molecular geometry (**Figure 17**). The non-bonding pair of electrons pushes the other bonds slightly upwards, and so the bond angles are now all less than 90° .

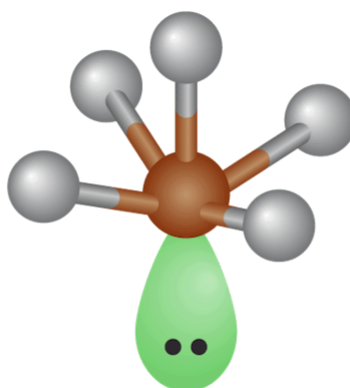


Figure 17. Square pyramidal.

 More information for figure 17

A 3D model showing a square pyramidal molecular geometry. The central atom is brown, with five bonded atoms in a square arrangement plus one additional non-bonding pair of electrons. The bonded atoms are connected to the central atom by brown bonds arranged at less than 90° angles due to the repulsion from the non-bonding pairs. The unshared electron pair is represented as a green teardrop shape positioned below the central atom, illustrating its effect on the bond angles.

[Generated by AI]

As **Figure 18** shows, four bonded pairs and two non-bonding pairs result in a square planar molecular geometry. Both the non-bonding pairs are found on the same axis, one above the plane and one below the plane. They exert equal forces on the atoms bonded in the other axis, their forces 'balance out' and so the bond angle stays at 90° .

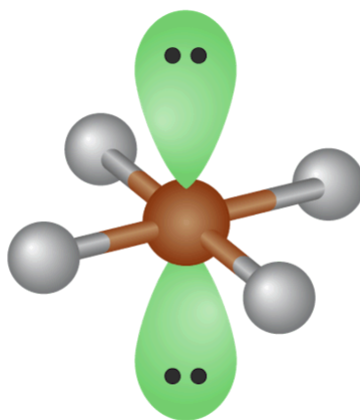


Figure 18. Square planar.

 More information for figure 18

The image illustrates a square planar molecular geometry. It shows a central atom bonded to four atoms arranged in a square plane. Two non-bonding electron pairs are depicted as larger green lobes, with one lobe situated above the plane and the other below. The non-bonding pairs are on the same axis, exerting equal forces that balance out, keeping the bond angles at 90° as explained in the accompanying text.

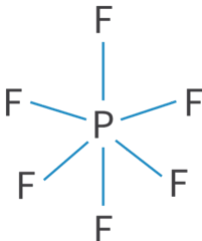
[Generated by AI]

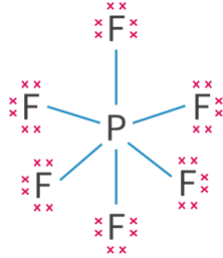
Worked example 3

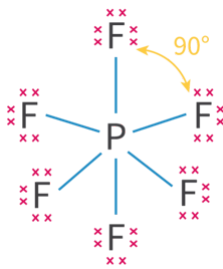
Draw the Lewis formula and use it to identify the molecular geometry and bond angles for the:

1. PF_6^- ion
2. BrF_5 molecule
3. XeF_4 molecule

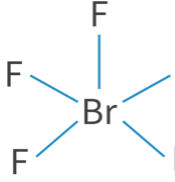
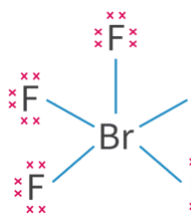
1. PF_6^- ion

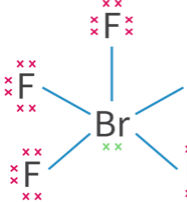
Solution steps	Calculations
Step 1: Calculate total valence electrons	$(P) + (F \times 6) + (\text{negative charge}) = (5) + (7 \times 6)$
Step 2: Identify the central atom. Here, it will be phosphorus as it is the least numerous (only one phosphorus compared to six fluorine), it is further left on the periodic table and is written first in the chemical formula. Join the peripheral atoms (fluorine) to the central	There are six fluorine atoms, which must be peripheral atoms so it can expand its octet. Thus, the structure of PF_6^- must 

Solution steps	Calculations
atom (phosphorous) with single bonds.	
Step 3: Add remaining electrons as valence electrons.	 The diagram shows a central phosphorus atom (P) bonded to five fluorine atoms (F) in an octahedral-like geometry. Each bond is represented by a blue line. Each fluorine atom has three lone pairs of electrons, represented by six red 'x' marks around each F. The phosphorus atom has five bonds, each contributing two electrons, for a total of 10 valence electrons around it, illustrating an expanded octet.
Step 4: Add double or triple bonds where necessary to ensure that all atoms fulfil octet rule. (Octet rule already achieved so not necessary.)	
Step 5: Check the number of electrons	

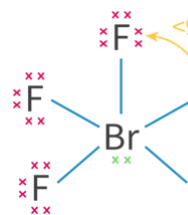
Solution steps	Calculations
added is correct as calculated in Step 1. Yes, 48 electrons are added in total.	
Step 6: Octet already expanded for phosphorus. This is possible as it's in period 3.	
Step 7: Identify molecular geometry and bond angles	Electron domains: 6 Bonded electron pairs: 6 Lone electron pairs: 0 Molecular geometry: Octahedral Angles: 90° 

2. BrF₅ molecule

Solution steps	Calculations
Step 1: Calculate total valence electrons	$(\text{Br}) + (\text{F} \times 5) = (7) + (7 \times 5)$ $= 42 \text{ total}$
Step 2: Identify the central atom. Here it will be bromine as it is the least numerous (only 1 bromine compared to 5 fluorine) and is written first in the chemical formula. Join the peripheral atoms (fluorine) to the central atom (bromine) with single bonds.	There are 5 fluorine atoms, which must be peripheral so it can expand its octet, so the structure of BrF_5 
Step 3: Add remaining electrons as valence electrons.	
Step 4: Add double or triple bonds where necessary to ensure that all atoms fulfil octet rule. (Octet rule	

Solution steps	Calculations
already achieved so not necessary.)	
Step 5: Check the number of electrons added is correct as calculated in Step 1. No, 40 electrons are added in total.	
Step 6: Octet already expanded for bromine. This is possible as it's in period 4. Octet must now be expanded further to accommodate an additional non-bonding pair of electrons.	

Solution steps	Calculations
Step 7: Identify shape and bond	Electron domains: 6 Bonded electron pairs: 5 Lone electron pairs: 1 Molecular geometry: Square pyramidal Angles: $<90^\circ$ (87°)



3. XeF₄ molecule

Solution steps	Calculations
Step 1: Calculate total valence electrons.	$\text{Xe} + (\text{F} \times 4) = 8 + (7 \times 4)$ $= 36 \text{ total}$
Step 2: Identify the central atom. Here, it will be xenon as it is the least numerous (only 1 xenon compared to 2 fluorine) and is written first in the chemical formula. Join the peripheral atoms (fluorine) to the central atom (xenon) with single bonds.	

Solution steps	Calculations
Step 3: Add remaining electrons as valence electrons.	
Step 4: Add double or triple bonds where necessary to ensure that all atoms fulfil octet rule. (Octet rule already achieved so not necessary.)	
Step 5: Check the number of electrons added is correct as calculated in Step 1. No, only 32 electrons are added in total.	
Step 6: An octet must now be expanded further to accommodate two additional non-bonding pairs of electrons. This is possible for the central atom xenon, as it's in period 5.	
Step 7: Identify shape and bond	Electron domains: 6 Bonded electron pairs: 4 Lone electron pairs: 2

Solution steps	Calculations
angles.	Molecular geometry: Square planar Angles: 90°



Study skills

Remembering the names of the molecular geometries can prove tricky if you have trouble visualising where the name has come from.

For the octahedral molecular geometry, the molecule we identified above will result in a shape with eight sides, known commonly as an octahedron.

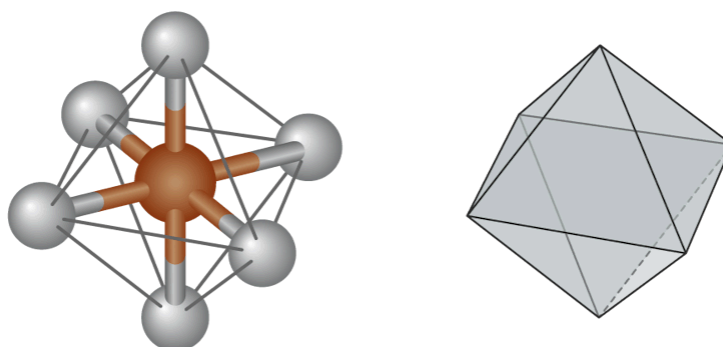


Figure 19. Octahedral.

 More information for figure 19

The image shows two main elements representing an octahedral molecular geometry. On the left is a 3D model of an octahedron with a central atom connected to six surrounding atoms. Thin lines indicate bonds between the central atom and the outer atoms, forming a symmetrical shape. The central atom is colored in a contrasting shade to the outer atoms, which helps to distinguish the core from the peripheries.

On the right is a geometric representation of an octahedron, illustrating its eight faces, twelve edges, and six vertices. The shape is depicted in a transparent manner, with solid and dashed lines highlighting the three-dimensional aspect of the octahedron. This part of the illustration aids in visualizing the regularity and symmetry of the octahedron geometry.

[Generated by AI]

Square pyramidal molecules can be linked to the four-sided shape, a pyramid, which we see in everyday life.

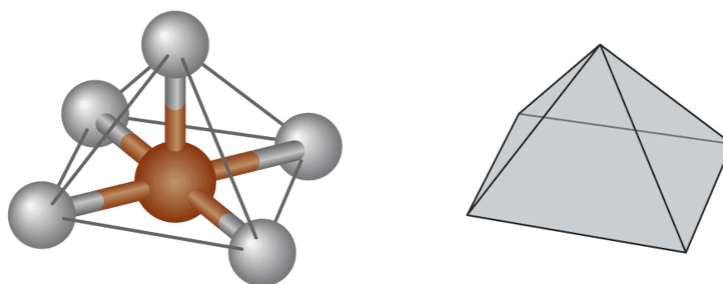


Figure 20. Square pyramidal.

 More information for figure 20

The image displays two elements: on the left, a square pyramidal molecular structure is depicted using spheres and sticks. A central sphere connects to four peripheral spheres arranged in a square, with an additional sphere at the vertex above the square plane, forming a pyramid shape. On the right, there is a three-dimensional geometric representation of a square pyramid, showing its base and triangular sides visually annotated.

[Generated by AI]

Square planar molecules are so called because they are four-sided structures, all existing on the same plane, that is, they appear flat.

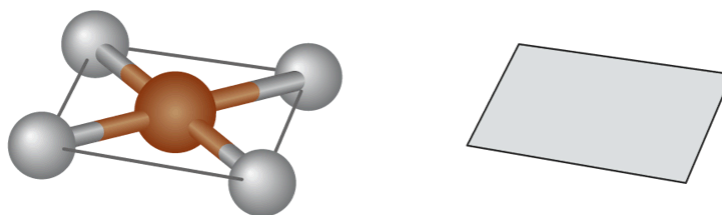


Figure 21. Square planar.

More information for figure 21

The image depicts a square planar molecular structure. It shows a central atom, represented as an orange sphere, connected to four peripheral gray spheres with lines, forming a square. These connections illustrate the spatial arrangement of atoms, where all atoms lie in the same plane, giving the molecule a flat, four-sided appearance. On the right side of the image, a shaded rectangle represents the plane on which these atoms exist, highlighting the planar nature of the molecular geometry.

[Generated by AI]

Summary of shapes

Table 1. Summary of VSEPR geometries for 2—6 elec

Number of electron domains	Electron domain geometry	Number of bonding electron domains	Number of non-bonding pairs of electrons	Molecular geometry
2	linear	2	0	linear

Number of electron domains	Electron domain geometry	Number of bonding electron domains	Number of non-bonding pairs of electrons	Molecular geometry
3	planar triangular	3	0	planar triangular

Number of electron domains	Electron domain geometry	Number of bonding electron domains	Number of non-bonding pairs of electrons	Molecular geometry
3	planar triangular	2	1	V-shaped

Number of electron domains	Electron domain geometry	Number of bonding electron domains	Number of non-bonding pairs of electrons	Molecular geometry
4	tetrahedral	4	0	tetrahedral
4	tetrahedral	3	1	pyramidal

Number of electron domains	Electron domain geometry	Number of bonding electron domains	Number of non-bonding pairs of electrons	Molecular geometry	
4	tetrahedral	2	2	V-shaped	

Number of electron domains	Electron domain geometry	Number of bonding electron domains	Number of non-bonding pairs of electrons	Molecular geometry	
5	triangular bipyramidal	5	0	triangular bipyramidal	

Number of electron domains	Electron domain geometry	Number of bonding electron domains	Number of non-bonding pairs of electrons	Molecular geometry
5	triangular bipyramidal	4	1	unsymmetrical tetrahedron/ seesaw

Number of electron domains	Electron domain geometry	Number of bonding electron domains	Number of non-bonding pairs of electrons	Molecular geometry	

Number of electron domains	Electron domain geometry	Number of bonding electron domains	Number of non-bonding pairs of electrons	Molecular geometry
5	triangular bipyramidal	3	2	T-shaped

Number of electron domains	Electron domain geometry	Number of bonding electron domains	Number of non-bonding pairs of electrons	Molecular geometry	
5	triangular bipyramidal	2	3	linear	
6	octahedral	6	0	octahedral	

Number of electron domains	Electron domain geometry	Number of bonding electron domains	Number of non-bonding pairs of electrons	Molecular geometry	
6	octahedral	5	1	square pyramidal	

Number of electron domains	Electron domain geometry	Number of bonding electron domains	Number of non-bonding pairs of electrons	Molecular geometry	
6	octahedral	4	2	square planar	

Number of electron domains	Electron domain geometry	Number of bonding electron domains	Number of non-bonding pairs of electrons	Molecular geometry

In section S2.2.4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/vsepr-id-45129/>) there was an activity to model different molecules on a macroscale using balloons to represent up to four electron domains surrounding a central atom. That understanding is built on now by creating models with up to six electron domains.

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:** Thinking skills — Designing procedures and models
- **Time required to complete activity:** 30 minutes
- **Activity type:** Group activity

Materials

- Modelling clay (plasticine) or marshmallows
- Q-tips (cotton buds) or toothpicks
- Pipe cleaners or thumbtacks

In this activity each Q-tip represents a bonded electron domain and each pipe cleaner can be bent to represent a non-bonding electron domain (**Figure 22**). Modelling clay can be used to represent the atoms being bonded together.

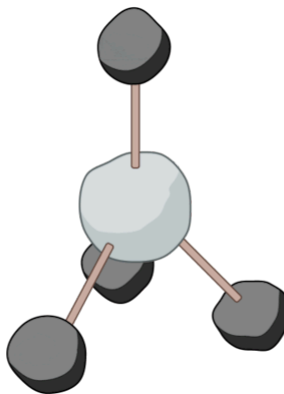


Figure 22. Using modelling clay and toothpicks to create a molecular model of methane, CH_4 .

 [More information for figure 22](#)

The image shows a three-dimensional molecular model of methane (CH_4). The central atom, representing carbon, is depicted using a larger piece of modeling clay in the center. Four smaller pieces of clay are attached around the central piece, each representing a hydrogen atom. These hydrogen atoms are connected to the central carbon atom using toothpicks, which symbolize the chemical bonds. The clay and toothpick model is arranged in a tetrahedral shape, common to methane, where the hydrogen atoms are positioned at the vertices and the carbon is at the center. This visual representation aids in understanding molecular geometry.

[Generated by AI]

Use these materials to create models of each of the molecular geometries associated with a central atom that has five or six electron domains. An example of a tetrahedral molecule, CH_4 , created with modelling clay and toothpicks is given for you in **Figure 23**.

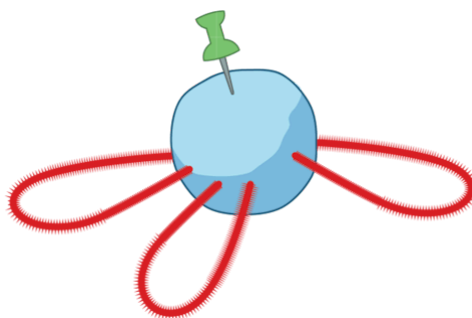


Figure 23. Non-bonding electron domains can be represented using pipe cleaners or thumbtacks.

 [More information for figure 23](#)

The image shows a model representing a tetrahedral molecular structure. A central sphere, made to resemble an atom, is at the center. Four red pipe cleaners extend outward from this central sphere in different directions, representing the bonds and electron domains. A thumbtack is inserted into the sphere, indicating a non-bonding electron domain. The pipe cleaners are bent to form loops, illustrating the spatial arrangement typical in tetrahedral molecules like CH_4 , ensuring maximum separation between the electron domains.

[Generated by AI]

Remember to orientate the electron domains in a position which gives them the most space relative to each other, similar to how electron domains do this in real life.

Things to consider

- How could different sized atoms be best represented?
- How should double and triple bonds be represented using the Q-tips?
- What would be the most effective way to demonstrate the additional repulsion caused by non-bonding pairs of electrons?
- Are there any other materials that would have been a better choice than what was suggested?